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Journal of Power Sources xxx (xxxx) 230760

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

Influence of metallic contaminants on the electrochemical and thermal behavior of Li-ion electrodes

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ARTICLE INFO

Keywords:
Lithium-ion battery recycling
Direct recycling
Contaminant
Isothermal microcalorimetry
Differential electrochemical analysis

ABSTRACT

Emerging nondestructive (direct) recycling techniques for lithium-ion batteries may introduce metallic impurities into recycled electrodes. In the present work, the impact of such nonionic contaminants on the practical performance of both anode and cathode materials is evaluated using a synergistic combination of electrochemical and thermal analysis. The impurities under study have been selected through evaluation of industrially shredded batteries, and include Fe^0, Al^0, Mg^0, Cu^0, and Si^0. The electrochemical behavior of materials containing each individual contaminant at either the anode or the cathode is evaluated in both half-cell and full-cell format. Further, the first-cycle thermal signatures of full cells are used to validate and complement electrochemical signatures, and the two techniques are used in conjunction to suggest distinct mechanisms of electrochemical reactivity for the various impurities. At the anode, metallic contaminants are found to disrupt performance through direct reaction with Li and may serve as weak catalysts to accelerate electrolyte degradation. At the cathode, metallic contaminants show evidence of crossover during formation cycling to disrupt SEI formation. We suggest that coupled electrochemical and thermal analysis may be used to both identify the presence of contaminants and to elucidate specific mechanisms of reactivity for metallic impurities under anodic and cathodic conditions.

1. Introduction

In an increasingly electrified society, lithium-ion batteries (LIBs) have risen to unmatched predominance in the secondary battery market. Growing demand for high-powered personal electronics, electric vehicles (EVs), and grid-scale storage solutions have fueled a nearly $40 billion LIB market – a value that is projected to nearly double by 2026 [1]. By 2028, global production of LIBs is anticipated to exceed 1200 GWh, or about 15 times the LIB capacity produced in 2016 [2]. The number of LIBs reaching their end of functional life will similarly burgeon, offset by the anticipated 8–10 year lifespan of the battery. By 2040, models predict that as many as 2.9 million packs (each containing hundreds to thousands of cells per pack) will be entering the waste stream each year from EVs alone [3]. Developing effective methods to recycle and recover the critical constituent materials of LIBs is crucial to both maintaining material supply chain stability [4] and mitigating environmental hazards associated with disposal.

There are several established approaches to LIB recycling, each relying on a distinct series of processing methods. Pyrometallurgical recycling uses high-temperature smelting to reduce LIB components to their elemental form, which may then be recovered and resynthesized into fresh components. Hydrometallurgical recycling relies on a series of leaching and separation processes to isolate components as salts at the molecular scale. Finally, emerging direct recycling methods seek to recover and refurbish components in nearly intact form, thereby retaining the maximum engineered value. In all reported LIB recycling operations, some form of mechanical tear-down (crushing, shredding) is used to access the internal battery components, reduce particle size, and facilitate downstream treatments [4]. This mechanical treatment, in conjunction with aqueous leaching and separation processes employed in hydrometallurgical recycling, is known to introduce metal impurities into the active materials [5–7]. These impurities may originate from the current collectors (Al, Cu); pack materials (Si, Mg); or the processing machinery itself (Fe).

Several previous studies have investigated these contaminants in the context of a pyrometallurgical or hydrometallurgical recycling scheme, where the metallic impurities are expected to be oxidized to dissolved ions. Unless otherwise removed, these ionic impurities pre-

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https://doi.org/10.1016/j.jpowsour.2021.230760
Received 26 July 2021; Received in revised form 15 October 2021; Accepted 9 November 2021
0378-7753/© 2021

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sent in the smelted or leached precursors may then be drawn into the synthesis process and substitute into the $Li(Ni_xCo_yMn_{1-x-y})O_2$ (NMC) lattice during resynthesis and sintering. This essentially results in a doping of impurity elements into the NMC, which has been proven to variably impact the performance of the cathode material. Several studies report the benefits of low-level impurity doping on cell performance, with reports of increased capacity retention, thermal stability, and rate performance through lattice stabilization [7–12]. However, inclusion of impurities in cathode resynthesis – particularly in high concentrations – is largely injurious to cell performance, due to morphological impacts on the secondary particle, increased cation mixing, and a reduction in specific capacity [10,11,13–15]. In virgin cathode manufacturing, the extent of doping, substitution, or coatings is an end-user specific choice, and thus the variable presence of impurities in recycled material is typically undesired.

While the study of ionic impurities in recycled NMC is relevant for existing pyrometallurgical and hydrometallurgical processes, these findings cannot be directly translated to instances where contaminants are anticipated to remain in a metallic (nonionic) form. This scenario is anticipated to practically arise in both direct recycling and in the reprocessing of manufacturing scrap. There has been limited work to date investigating the effect of metallic contaminants on battery active material performance. However, the existing literature implies that these impacts may be significant. It has been suggested that larger metallic impurities may physically puncture the separator [16], while smaller metallic impurities may diffuse through the porous separator and migrate to the counter electrode [9,17], leading to internal short circuit in both instances. Inclusion of metallic Al in powder and foil form has been shown to reduce both capacity and Coulombic efficiency of NMC [17], possibly through the surface oxidation of aluminum that impedes Li diffusion. Addition of powdered metallic Cu to NMC has been shown to induce electrochemical failure, ostensibly through the initiation of Li dendrite growth following Cu migration to the anode [9]. There are no existing reports on the impacts of other relevant contaminants (Fe, Mg, Si) in metallic form on cathode performance. Further, the anode has been largely overlooked in existing studies of recycled materials, given the lower material value of graphite as compared to Co and Mn. There are no known reports investigating impurities relevant to recycling and remanufacturing – either ionic or metallic – intentionally introduced into the anode. Finally, the phenomenon of crossover, whereby dissolved transition metals originating from one electrode migrate and are redeposited at the other electrode, has been well-documented [18], and is anticipated to substantially impact the performance of impurity-containing recycled materials.

In assessing the impacts of contaminants on cell performance, *in situ* approaches have generally been limited to electrochemical analysis. Electrochemical techniques enable practical measurement of cell capacities and Coulombic efficiencies, and manipulation of electrochemical data (dQ/dV; CV; EIS) may yield insights regarding physicochemical changes and mechanisms of reaction. However, to glean a complete understanding of what is inherently a chemo-mechanical-electrochemical system, complementary analytical techniques are critical. In previous impurity studies, *in situ* electrochemical data has been coupled with complementary *ex situ* techniques, such as XRD, SEM, and TEM [5]. Only one known study paired *in situ* electrochemical analysis of electrode contaminants with another *in situ* technique. In this report, infrared (IR) thermography was used as a quality control metric to identify regions of electrodes containing metallic defects, which were observed to have reduced emissivity (i.e., more rapid heat dissipation) relative to pristine sites [17]. Such nondestructive, *in situ* analysis of contaminant effects is critical to refining a mechanistic understanding of impurity effects on electrode functionality.

Isothermal microcalorimetry (IMC) is a powerful *in situ* technique by which the thermal response of a cell may be continuously measured during cycling. In the context of intercalation chemistries, IMC has notably been utilized to measure parasitic reactions during cycling [19] [–] [21] and to decouple the Faradic and non-Faradic contributions to cell heating [20–22]. Standard microcalorimetry techniques have previously been applied to the analysis of coin-type cells, which may not accurately represent behavior in larger cell formats (i.e., pouch cells). To that end, we have developed and herein report results from a highly sensitive isothermal microcalorimeter, capable of detecting and quantifying thermal signals as low as 1 mW from pouch cells. Further, our calorimeter design enables deconvolution of heat produced at the anode and at the cathode – a powerful capability for the analysis of practical full cells. Our findings suggest that thermal analysis may be used both to complement electrochemical analysis and as a screening tool to selectively identify impurities in recycled or reprocessed LIB materials.

The present work offers an analysis of the impacts of a variety of metallic contaminants (Al, Cu, Fe, Mg, Si) on both cathode and anode behavior. The choice of these specific impurity elements were based on chemical analysis of industrially generated shredded battery materials (hereafter "black mass"). Using electrochemical methods with microcalorimetric analysis, we highlight the synergy of these two techniques for the detection and specific identification of trace-level contaminants in recycled materials. Further, this coupled approach reveals specific mechanisms of electrochemical reactivity for each contaminant, which may not be evident through each technique in isolation. Taken together, this analysis provides a novel approach to identifying metallic impurities and suggests which metallic contaminants may be most problematic in the context of recycled LIB electrodes.

2. Materials &

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2.1. Industrial black mass: acquisition & characterization

Shredded battery materials (black mass) originating from Li-ion packs and cells was acquired from four industrial suppliers. Processing conditions and shredding techniques vary between suppliers with specifics largely proprietary. Size fractionation via sieving was conducted by Suppliers 1 and 2. Sample designations and processing parameters, where known, are presented in Table 1.

Characterization of the as-received black mass samples included optical imaging and scanning electron microscopy/energy dispersive X-ray spectroscopy (SEM/EDS; Phenom XL Benchtop SEM). Inductively coupled plasma mass spectrometry (ICP-MS) was conducted using a NexION 350D spectrometer (PerkinElmer) with an Opalmist nebulizer. Prior to ICP-MS analysis, NMC samples underwent thermal treatment (to remove residual carbon) and acid digestion. Specifically, triplicate samples were weighed to 100 ± 0.1 g and loaded in an acid-cleaned quartz crucible. Samples were then heated at 700 °C for 2 h in a muffle furnace. Following addition of HCl and HNO_3 (trace metal grade), samples were transferred to an acid-cleaned Teflon microwave reaction vessel, and subject to a microwave reaction program. Once cool, samples were transferred to polypropylene tubes (trace metal grade) and diluted to 50 mL total volume with 18.2 MΩ deionized water. A blank sample

Table 1
Identification of shredded black mass samples obtained from industrial suppliers.

Sample ID	Supplier	Processing Conditions *(Wet vs Dry)*	Shredding Method	Size Fraction
S1	1	Wet	Hammer Mill	>63 μm
S2	1	Wet	Hammer Mill	<63 μm
S3	2	Dry	Shear Shredder	63–350 μm
S4	2	Dry	Shear Shredder	<63 μm
S5	3	Dry	Hammer Mill	–
S6	3	Dry	Hammer Mill	–
S7	4	Dry	Shear Shredder	–

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was subject to the complete digestion and analysis protocol to correct for background and any potential interference.

2.2. Electrode fabrication

All electrodes and cells utilized in this study were produced at the U.S. Department of Energy's (DOE) CAMP (Cell Analysis, Modeling and Prototyping) Facility, Argonne National Laboratory, and were prepared in a dry room operating below −40 °C dew point.

The metallic impurities under study included iron (Fe^0: 10 μm, 99.9% purity, Aldrich); aluminum (Al^0: 5 μm, 99.5% purity, Aldrich); magnesium (Mg^0: 325 mesh (~44 μm), 99% purity, ACROS); copper (Cu^0: 5 μm, 99.8% purity, Goodfellow); and silicon (Si^0: 150 nm, 99.5% purity, Paraclete Energy).

Negative electrodes (anodes) were comprised of graphite powder (Superior Graphite, SLC1520P); carbon black (Timcal C45); binder (Kureha 9300); and oxalic acid (adhesion promotor). Baseline (impurity-free) anodes contained 91.83 wt% graphite, 2 wt% carbon black, 6 wt% binder and 0.17 wt% oxalic acid. Selected impurities were added in metallic form at 1 wt%, substituting for the graphite. Thus, impurity-containing anode compositions were 90.83 wt% graphite, 1 wt% impurity, 2 wt% carbon black, 6 wt% binder, and 0.17 wt% oxalic acid. Metal impurities studied in the anode included Fe^0, Al^0, Mg^0, and Cu^0.

Positive electrodes (cathodes) were comprised of NMC-111 power (Toda, NM-3101); carbon black (Timcal C45); and PVDF binder (Solvay, 5130). Baseline (i.e., impurity-free) cathodes contained 90 wt% NMC-111, 5 wt% carbon black, and 5 wt% binder. Selected impurities were added in metallic form at 1 wt%, substituting for the NMC. Thus, impurity-containing cathode compositions were 89 wt% NMC-111, 1 wt% impurity, 5 wt% carbon black, and 5 wt% binder. Metal impurities studied in the cathode included all of those studied in the anode (Fe^0, Al^0, Mg^0, Cu^0) as well as Si^0.

Mixtures (slurries) with the noted compositions were prepared with a Thinky ARE-310 mixer. For all anodes, the binder solution, carbon black and any metal impurity were first mixed together with two zirconia milling balls (3 min at 2000 rpm followed by a 30 s degas at 2200 rpm; 10 mm mixing ball diameter). The graphite powder was then mixed in, followed by the oxalic acid. A final mixing step was then performed to ensure homogeneity. For all cathodes, the binder solution, carbon black and any metal impurity were first mixed together according to the previously described parameters. Half the amount of the NMC-111 powder and additional NMP was then added, and the solution was remixed. The rest of the cathode powder was added and the solution again remixed, with NMP added and a final mix performed to adjust the viscosity of the slurry.

Anode and cathode slurries were cast using a reverse comma coater (A-Pro; 0.5 m/min; 50 mm coating width). Anodes were coated onto battery-grade Cu foil (LS Mtron; 10 μm) and cathodes were coated onto battery-grade Al foil (Dong-Il Aluminum Co., Ltd; 20 μm). Total anode loading (active material + carbon black + binder) for all electrodes was ~6.4 mg/cm^2 (1.9 mAh/cm^2); total cathode loading was ~11.3 mg/cm^2 (1.6 mAh/cm^2). After coating, all electrodes were roll pressed to a target porosity of 35% using a heated roll press (A-Pro; 80 °C at 0.5 m/min).

2.3. Cell fabrication

The prepared electrodes were assembled into CR2032 coin cells for electrochemical analysis. For half coin cells, an electrode disc (15 mm anode or 14 mm cathode) was paired with a 15.6 mm disc of Li metal (MTI Corporation). Half cells were prepared using a 16 mm disc of Celgard 2320 separator (20 μm thickness) and ~90 μL of Gen2 electrolyte (1.2 M $LiPF_6$ in ethylene carbonate (EC): ethylmethylcarbonate (EMC), where EC:EMC = 3:7 wt%. For full coin cells, a 14 mm cathode was paired with a 15 mm anode. For full-cell testing of cathode contaminants, a baseline (i.e., impurity-free) anode was used as the counter-electrode; for full-cell testing of anode contaminants, a baseline cathode was used as the counter-electrode. Full cells were prepared using a 5/8" diameter disc of Celgard 2320 separator (20 μm thickness) and 29 μL of Gen2 electrolyte. Electrolyte was purchased from Tomiyama Pure Chemical Industries.

Single-layer (SL) xx3450 pouch cells were assembled for thermal analysis using the prepared electrodes and Celgard 2320 separator (20 μm thickness). Cathodes were punched to a coated area of 14 cm^2 (31.44 mm × 45.0 mm) and anodes were punched to a coated area of 14.9 cm^2 (32.4 mm × 46.0 mm), such that the cathode was the limiting electrode and the anode was oversized. Cells were sealed dry (i.e., without electrolyte) under vacuum, with electrolyte added just prior to thermal analysis, as described in Section 2.5.

2.4. Electrochemical cycling

Electrochemical cycling was conducted on half and full coin cells using a multichannel cycler (Maccor). For all cells, formation consisted of 4 cycles of a C/10 constant-current (CC) discharge followed by a C/10 CC charge, with a 2 min rest period between steps. Anode half cells were cycled between 0.005 and 1.5 V; cathode half cells were cycled between 3.0 and 4.3 V; full cells were cycled between 3.0 and 4.2 V. For half cells, long-term cycling (following formation) consisted of 50 cycles at C/3 CC charge and discharge, with a 2 min rest period between steps. Anode half cells were cycled between 0.005 and 1.5 V; cathode half cells were cycled between 3.0 and 4.3 V. For full cells, long-term cycling (following formation) was conducted between 3.0 and 4.2 V. The cycling protocol consisted of a total of 255 cycles, which included five repetitions of the following series of steps: 1) C/20 discharge to 3.0 V, followed by a 10 min rest period; 2) 1 cycle of C/20 CC charge and C/20 discharge, with a 10 min rest period between steps; 2) 48 cycles of C/3 CC charge with a C/5 trickle charge and C/2 discharge; 3) 1 cycle of 1C CC charge (C/5 trickle charge) and 1/C discharge, with a 2 min rest period between steps; 4) 1 cycle of 1C charge (C/5 trickle charge; preparation for Hybrid Pulse Power Characterization (HPPC)); 5) 9 sets of 10-s HPPC pulses, consisting of 5 C discharge followed by 3.75 C charge.

2.5. Thermal analysis

Thermal analysis was conducted using a microcalorimeter designed and fabricated in-house to precisely monitor heat generation from small-format pouch cells. A schematic of the microcalorimeter setup is provided in Fig. A1. Cells were sandwiched between two custom-designed Cu heat spreaders (busbars), with thermal grease applied between the cell and each busbar for enhanced thermal contact. This stack was compressed to 20 psi via calibrated springs. Cell tabs were clamped separately to each busbar to provide electrical contact. Four thermoelectric generators (TEGs; Custom Thermoelectric) were arranged in series on each busbar, with the "hot" side of each TEG oriented towards the battery and the "cold" side oriented outwards. The entire unit was sealed in an Al heat sink, which was then submerged in an isothermal Fluke calibration bath (50 : 50 by volume ethylene glycol: deionized water; stability of ±0.0007 °C at 25 °C). Shielded twisted pair cables, selected to minimize electrical noise, were soldered to the Cu busbars and routed out of the microcalorimeter using liquid-tight pass-through ports.

Prior to analysis, the microcalorimeter was calibrated using a thin Kapton resistive heater, which was installed in the cell site. Six-point calibration was achieved by passing a series of wattages (5 mW–20 mW; Keysight power supply) through the resistive heater, and measuring the TEG voltage output (Keysight Data Acquisition Sys-

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tem 34970A). Linear regression was applied to obtain a calibration curve, which was employed to analyze thermal data (described below).

Immediately prior to thermal analysis, pouch cells produced at the CAMP facility (described in Section 2.3) were opened under Ar atmosphere, filled with 470 μL of Gen2 electrolyte, and re-sealed under Ar. Following installation in the microcalorimeter, the cells were rested for 12 h, which was determined to be the minimum time required to achieve thermal equilibrium. Cells were then subjected to the full-cell formation protocol described in Section 2.4 using a multichannel cycler (Arbin). Electrochemical data was collected via both the Arbin cycler and the Keysight DAQ; thermal data was collected using the Keysight DAQ.

Thermal data was processed using software and analysis techniques developed in-house. Anode and cathode TEG signals were first baseline corrected using the average value measured at thermal equilibrium. A custom smoothing filter ("J-Filter") based on a factored difference between raw signals and an imposed Savitsky Golay filter was applied to mitigate data noise, without compromising peak resolution. In the following report, cell heat output is presented in a voltage-dependent form (W).

3. Results

3.1. Identification of industrially relevant metallic impurities

To develop a system of study with relevance to an industrial recycling/reprocessing line, black mass (shredded battery material) was obtained from four different suppliers (Table 1). Both optical (Fig. A2) and EDS (Fig. 1) images show evidence of impurities in the shredded material. The most common impurity elements reported in literature (Al, Cu, Fe, Si) were observed via EDS; trace levels of P, K, and S were also present in several samples. To quantify impurity quantities, the various black mass samples were analyzed via ICP-MS (Table 2). There is significant heterogeneity in impurity content across suppliers, reflecting both the diversity in shredding and processing techniques utilized and the variety of battery materials analyzed. However, across all samples, Al and Cu – ostensibly from the current collector – are found to be the dominant impurity elements. Fe, Si, and Mg are detected at much lower levels (0.01–1.6 wt %), with measured weight fraction decreasing as Fe ≤ Si < Mg across all samples.

The samples from Suppliers 1 and 2 (S1, S2 from Supplier 1; S3, S4 from Supplier 2) had been size fractionated via sieving prior to acquisition. Samples S1 and S3 contained material from the larger size fraction (>63 μm in each case), while samples S2 and S4 contained material from the smaller size fraction (<63 μm). In both cases, the larger size fraction was found to contain significantly higher levels of Al and Cu

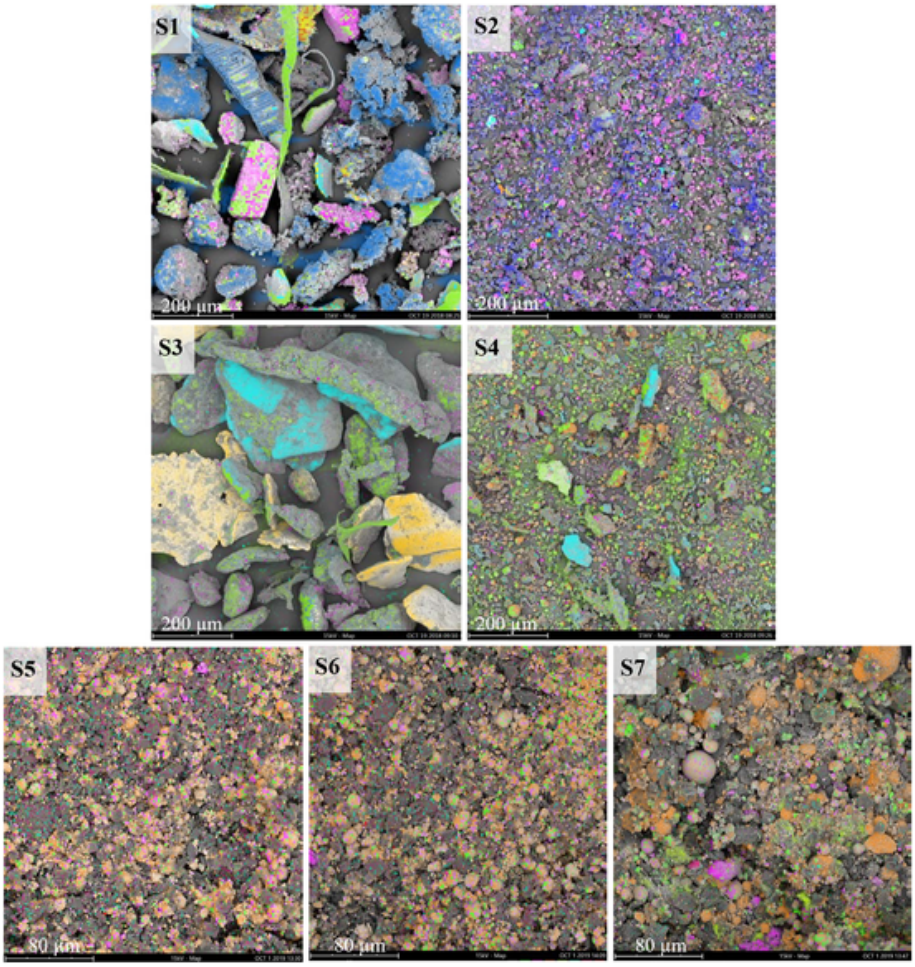

Fig. 1. SEM/EDS data of shredded black mass samples obtained from industrial suppliers. Sample IDs correspond to designations assigned in Table 1. Colors correspond to elements present: **S1**: O = green; Ni, Co = purple; Mn = orange; Al, Si = light blue; Fe, Mg = yellow; P = teal; K = dark blue; **S2:** O = dark blue; Ni = green; Co = purple; Mn = orange; Al, Si = light blue; Fe = yellow; P = teal; **S3–S4:** O = green; Ni, Co = purple; Mn = orange; Al = light blue; Cu = gold; P = teal; **S5–S7:** Ni = green; Co = purple; Mn = orange; Al = yellow; P = teal; S = magenta; Si = red. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

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Table 2
Summary of elemental analysis of industrial black mass samples (ICP-MS). Concentration is reported as a weight percentage of the full sample, with the balance accounted for by standard cathode active material elements (Ni, Mn, Co, O).

Impurity Element[a]	Concentration (wt%), By Sample ID						
	S1	S2	S4	S5	S6	S7	S8
Al	4.3	1.7	16.0	6.6	0.03	0.04	3.07
Cu	1.1	0.6	15.7	3.2	0.55	0.61	1.44
Fe	1.6	0.5	0.4	0.5	0.06	0.07	0.10
Si	1.6	0.4	0.03	0.1	0.02	0.02	0.04
Mg	0.5	0.2	0.01	0.02	0.01	0.01	0.02

[a] Cathode active material elements (Ni, Mn, Co, O) comprise the balance of sample mass.

impurity. For Supplier 1, the larger size fraction contained ~2.5 times the amount of Al impurity and ~1.8 times the amount of Cu impurity as the smaller fraction; for Supplier 2, a similar trend was observed (~2.4 times higher Al and ~4.9 times higher Cu in the larger fraction). Notably, the two suppliers used different mechanical treatment methods: Supplier 1 utilized hammer milling, while Supplier 2 utilized shear shredding. This implies that impurity content in industrial black mass may be more highly dependent on downstream processing (the details of which were proprietary for the materials under study) than on the specific mode of mechanical treatment.

This analysis of industrial black mass informed the design of model systems to investigate the individual impacts of each metallic contaminant on both anode and cathode behavior. In particular, each of the five impurity elements identified in Table 2 (Al, Cu, Fe, Si, and Mg) were studied at a 1 wt% impurity level, which falls within the range of what may practically be anticipated from an industrial recycling or remanufacturing process. Maintaining a constant level of contamination across all samples allows for the direct comparison of specific electrochemical and thermal signatures associated with each impurity. These signatures can then serve to selectively identify the presence of individual contaminants in recycled material.

3.2. *Impact of metallic impurities on anode performance*

The electrochemical signatures of graphite anodes containing four industrially relevant metallic impurities (Fe^0, Al^0, Mg^0, and Cu^0) have been evaluated in both half-cell (CR2032 coin cells; graphite vs Li) and full-cell (pouch cell; graphite vs NMC-111 cathode) formats to assess the impact of each metallic impurity on anode performance. Thermal signatures of contaminated anodes have also been evaluated in full cells. Half cells enable isolation and identification of processes occurring at the anode, which informs a mechanistic understanding of the electrochemical reactivity induced by the presence of contaminants. Full cell testing evaluates the effect of anode contamination in a practical cell format, which is highly relevant in the context of directly recycled materials.

The impacts of metallic contaminants may be most strongly observed during the initial formation cycles, although several contaminant chemistries continue to impact electrochemical performance throughout extended cycling (Figs. A3-A4). Thus, in the present report, emphasis will be placed on analysis of electrochemical and thermal signatures during formation cycling. Specific impacts of each metallic contaminant on the electrochemical and thermal performance of the anode are highlighted, with the aim of both offering a fundamental mechanistic explanation of reactivity and identifying diagnostic features that may be employed to screen for impurities in recycled materials.

Differential capacity plots (dQ/dV) of half cells with impurity-containing anodes, as compared to pristine anodes, are shown for the first four formation cycles in Fig. 2; analogous dQ/dV plots for full cells with impurity-containing anodes are shown for formation cycles in Fig. 3. Differential capacity analysis in full cells results in a convolution of anode and cathode behaviors, which must be decoupled to identify contributions associated with each electrode. Previous reports have undertaken this decoupling via curve-fitting algorithms [23,24], suggesting typical peak positions that may be linked to anodic or cathodic behavior. The present study will utilize these interpretations, as well as comparison with half-cell differential capacity plots, to identify electrochemical changes attributable to each electrode contaminant. It should be noted that the electrochemical performance of cells containing anode impurities varied significantly between cell replicates, perhaps suggesting inhomogeneity in the dispersion of metallic contaminants throughout the electrode. Where relevant, two dQ/dV plots have been included for each cell condition (reflecting two replicates) to more accurately reflect the scope of observed behavior.

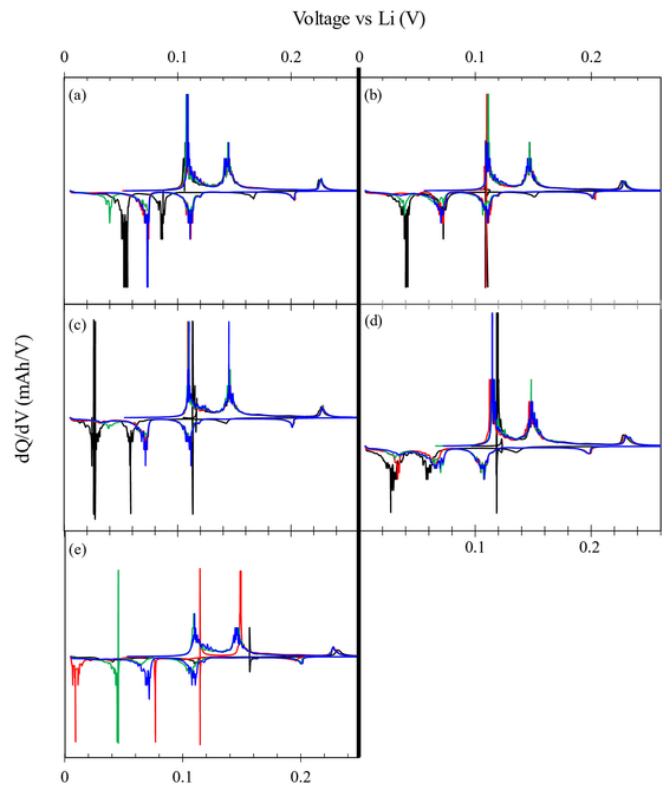

Fig. 2. Differential capacity (dQ/dV) during formation cycles for anode half cells (graphite anode vs Li_0) prepared with metallic impurities: (a) baseline (no impurity); (b) Fe impurity; (c) Al impurity; (d) Mg impurity; (e) Cu impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Finally, thermal signatures for full cells containing metallic anode impurities have been acquired. In Fig. 4, the thermal signatures have been normalized by the signature of a baseline cell so that behavior uniquely attributable to each impurity may be differentially distinguished. Raw thermal signals for each cell are given in Fig. A5.

Each metallic anode impurity will be taken up in detail in the following sections.

3.2.1. *Fe^0-contaminated anode*

The presence of metallic Fe^0 at the anode is generally found to have minimal electrochemical impact. As shown in Fig. 2(b), a slight shift in first-cycle reductive behavior is observed in half-cell format: Specifically, reductive peaks associated with the Fe^0-containing anode are found to shift to lower voltages as compared to the baseline cell, implying reduced thermodynamic favorability. The reductive peak occurring at 0.167 V in the baseline cell is shifted to 0.151 V in the Fe-